

The University of Akron

Department of Statistics

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Decision Model for Telecom Business – Final Project

Course Name: Decision Tree

Instructor: Mark Fridline

Project Title: Decision Tree Model for Effective Management of Telecommunication Customer Churn

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“If your retention is poor then nothing else matter.”

-Brain Balfour

Project Background:

In the highly competitive telecommunications industry, customer retention is just as important as customer acquisition. One of the most significant challenges faced by telecom companies is customer churn the loss of clients or subscribers to competitors. Churn directly impacts revenue, customer lifetime value, and long-term growth.

As telecom companies manage millions of customers, it becomes essential to predict which users are at risk of leaving and understand why. Accurate churn prediction allows providers to take proactive steps to improve retention through targeted strategies.

This project focuses on developing and comparing decision tree-based models to identify key factors that influence customer churn. Decision trees offer a simple yet powerful visual representation of decision rules, making them ideal for classification tasks like churn analysis. By analyzing historical customer data, the models aim to uncover patterns and provide actionable insights that telecom companies can use to reduce churn and improve customer loyalty.

Objective:

The objective of this project is to develop classification models that can accurately predict whether a customer will churn based on their service usage patterns and account details. These models will support telecom companies in making data-driven decisions to improve customer retention strategies.

By applying various decision tree algorithms and logistic regression, we aim to:

- Identify the most influential factors associated with customer churn.
- Compare the performance of different predictive modeling techniques.

- Select the most effective model based on accuracy, interpretability, and practical business value.

Ultimately, the goal is to provide a reliable framework that can help minimize customer attrition and support strategic decision-making.

Dataset Description:

The dataset used in this project was sourced from Kaggle and contains customer data from a telecommunications service provider. It includes detailed demographic, account, and usage information for each customer, making it well-suited for churn prediction modeling.

- Total Observations: 7,043 customers
- Total Variables: 21
- Response Variable: Churn (Binary – Yes or No)

Key Predictor Variables:

- Contract Type (Month-to-month, One year, Two year)
- Payment Method (Electronic check, Mailed check, Credit card, Bank transfer)
- Internet Service (DSL, Fiber optic, None)
- Monthly Charges and Total Charges
- Tenure (Number of months a customer has stayed)
- Partner, Dependents, Senior Citizen, Phone Service, Online Security, etc.

The dataset contains both categorical and numerical variables, requiring careful handling during preprocessing and modeling. The target variable Churn is imbalanced, with approximately 26.5% of customers marked as "Yes" (churned), and 73.5% marked as "No" (not churned), which may influence model performance and interpretation.

Tools & Techniques:

To analyze the dataset and build predictive models, we used the following tools and techniques:

Software:

- IBM SPSS – Used for data cleaning, exploratory analysis, and modeling

Modeling Techniques:

- We applied four classification methods to predict customer churn:
- CHAID (Chi-squared Automatic Interaction Detector):

  A decision tree algorithm that splits nodes based on chi-square tests for independence, ideal for categorical variables.
- Exhaustive CHAID:

  An enhanced version of CHAID that performs an exhaustive search to find the best splits, potentially resulting in deeper and more refined trees.
- CART (Classification and Regression Trees):

  A binary-splitting decision tree algorithm that uses the Gini index or entropy to create interpretable tree structures. Pruning is applied to avoid overfitting.
- Logistic Regression:

  A statistical model that estimates the probability of a binary outcome and quantifies the relationship between predictor variables and churn likelihood.
- Each model was evaluated using key performance metrics such as Risk Estimate, Average Squared Error (ASE), Sensitivity, and Specificity.

Exploratory Data Analysis:

Frequency Table

Whether the customer churned or not

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Not Churned	5174	73.5	73.5	73.5
	Churned	1869	26.5	26.5	100.0
	Total	7043	100.0	100.0	

Fig 1: Frequency Table of Churn Variable

The dataset consists of 7,043 total observations. Of these, 1,869 customers (26.5%) had churned and 5,174 customers (73.5%) had not. This demonstrates a notable class imbalance, with the majority of customers not churning. The skewed distribution can impact model performance by making it more difficult for certain algorithms to correctly classify the minority class (churned customers), especially if not properly addressed during model training or evaluation.

Gender of customer

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Female	3488	49.5	49.5	49.5
	Male	3555	50.5	50.5	100.0
	Total	7043	100.0	100.0	

Fig 2: Frequency Table of Gender Variable

The dataset contains 3,488 female customers (49.5%) and 3,555 male customers (50.5%), indicating an almost perfectly balanced gender distribution. This suggests that gender is unlikely to be a major factor in churn prediction, as there is no clear dominance of one gender group over the other.

Descriptive Statistics:

Descriptives

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Number of months the customer has stayed with the company	7043	0	72	32.37	24.559
The amount charged to the customer monthly	7043	18.25	118.75	64.7617	30.09005
The total amount charged to the customer	7032	18.80	8684.80	2283.3004	2266.77136
Valid N (listwise)	7032				

Fig 3: Descriptive statistics of quantitative variables

The descriptive statistics provide key insights into the distribution and range of numerical variables:

- Tenure (Number of months the customer has stayed):

  The average customer stays for 32.37 months, with a maximum of 72 months. This variable may be a strong indicator of customer loyalty, as longer-tenured customers are generally less likely to churn.

- Monthly Charges:

  Charges range from $18.25 to $118.75, with an average of $64.76 and a standard deviation of 30.09, suggesting significant variation in service plans and billing.

- Total Charges:

  The total amount charged across a customer's lifecycle ranges from $18.80 to $8,684.80, with an average of $2,283.30. The high standard deviation (2,266.77) implies diverse usage patterns and billing histories.

These variables were considered crucial for modeling and interpreting customer churn behavior.

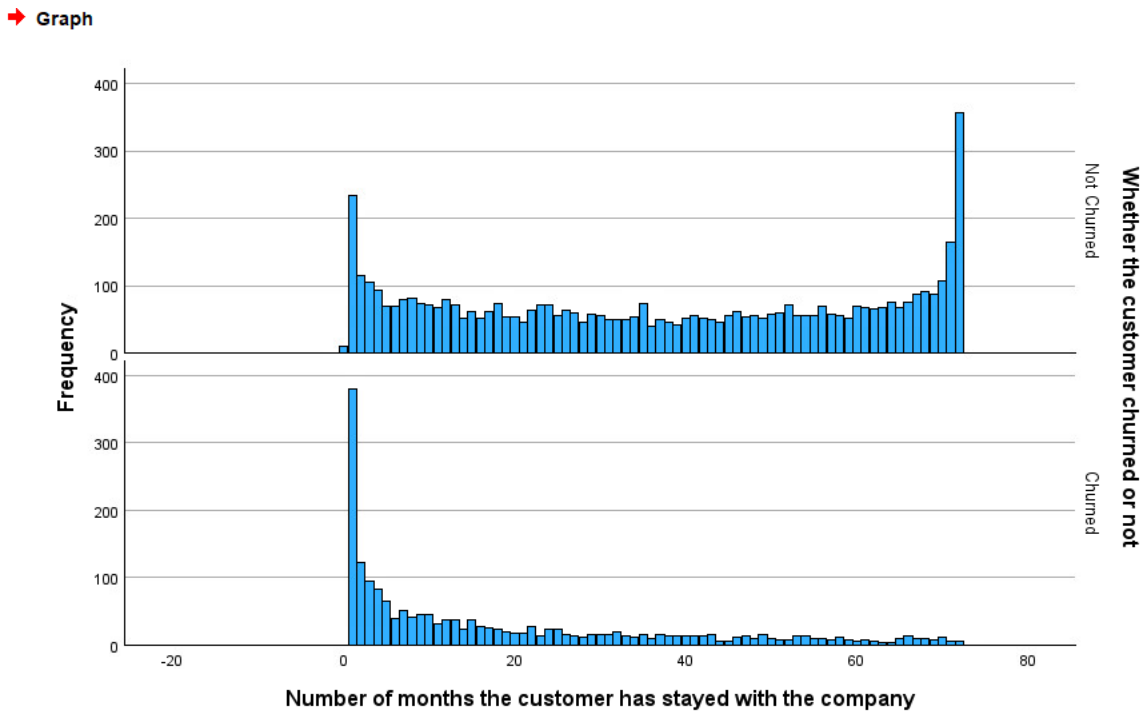


Fig 4: The Histogram of number of month customer stayed with churned status

This histogram clearly shows a relationship between tenure and churn status:

- Customers with short tenures are significantly more likely to churn. The bottom half of the chart, representing churned customers, shows a steep drop-off in frequency after the first 10–15 months.
- As tenure increases, churn likelihood decreases. The top half of the chart shows that non-churned customers are more likely to have stayed with the company for 60+ months.
- Notably, over 350 customers who stayed more than 60 months did not churn, while fewer than 20 customers who churned had reached that same tenure level.

This suggests that customer loyalty increases significantly with time, making tenure a key predictor for churn in future models.

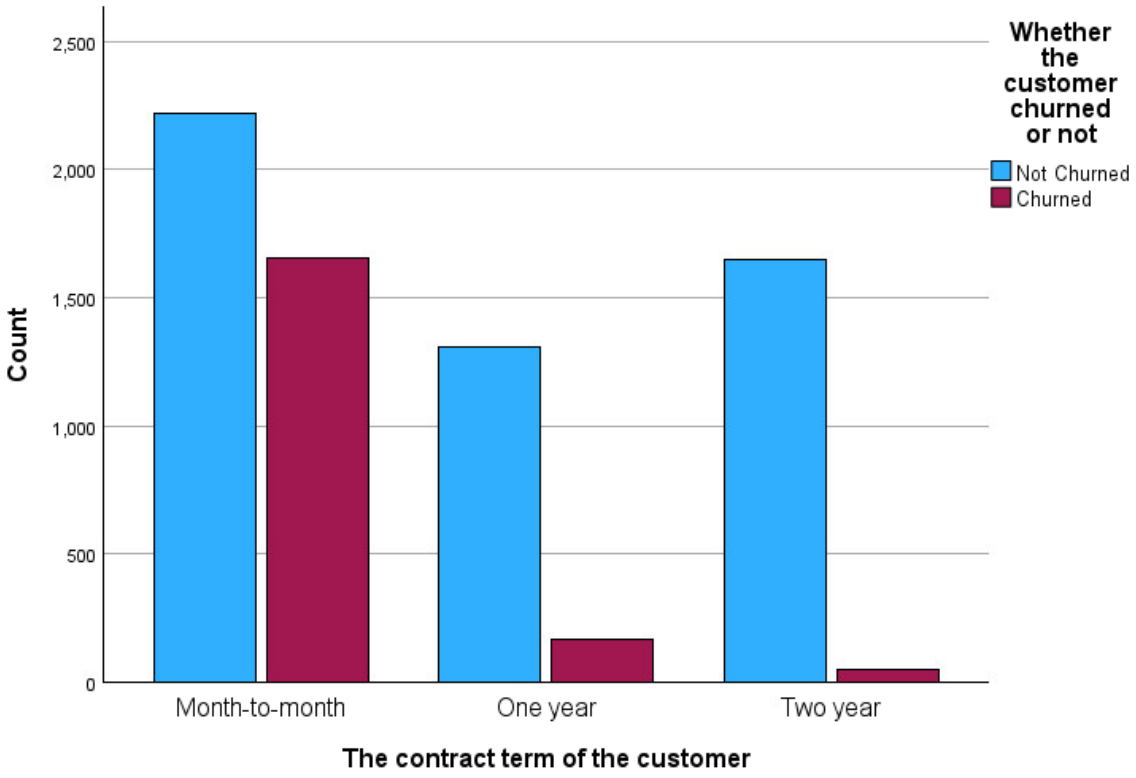


Fig 5: Bar Chart for contract term of customer separated with Churn status

This bar chart illustrates the strong association between contract type and customer churn:

- Customers with month-to-month contracts had the highest churn count, with 1,500+ customers leaving under this arrangement.
- In contrast, churn was significantly lower for customers on one-year and especially two-year contracts, where fewer than 100 customers churned.

These findings suggest that contract duration is a key predictor of customer retention. Customers who commit to longer-term contracts are more likely to stay, possibly due to financial incentives, contract obligations, or greater satisfaction with the service.

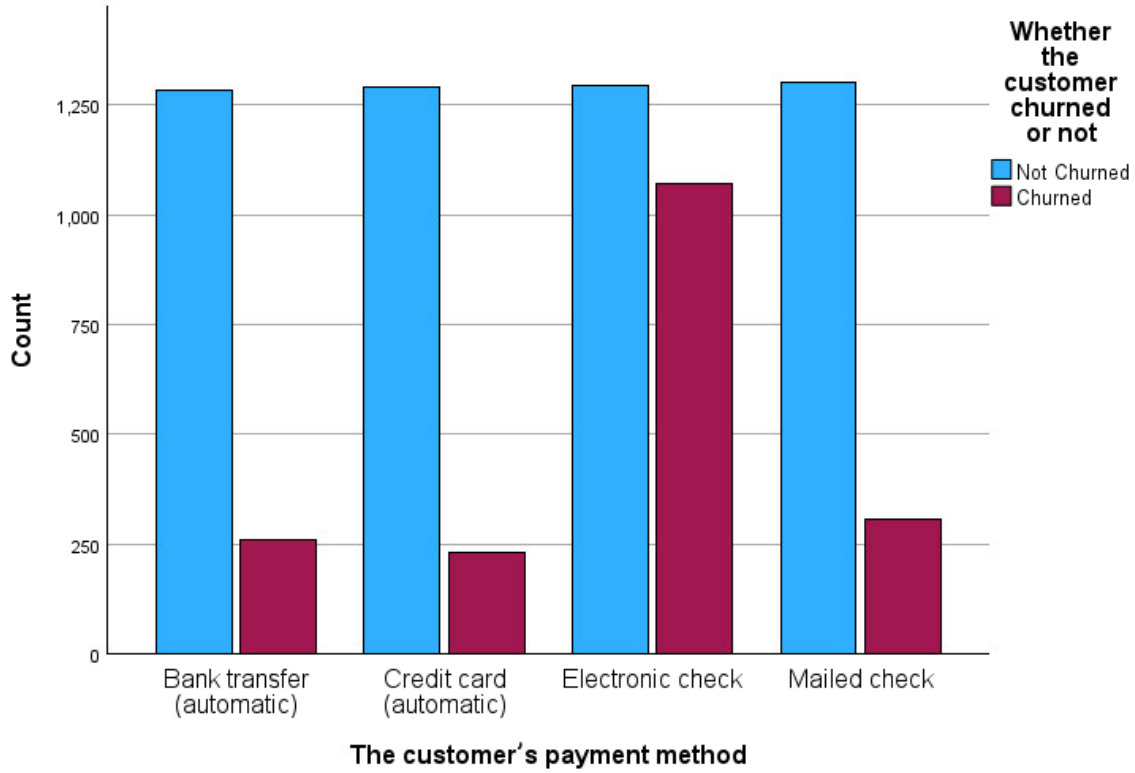


Fig 6: Bar Chart of Payment Methods separated with churn variable

This chart highlights a clear relationship between payment method and churn behavior:

- Customers using electronic checks had the highest churn count, with over 1,000 churned customers (significantly more than any other method).
- In contrast, customers using automatic credit card payments had the lowest churn rate, with fewer than 250 churned cases.
- Both bank transfer and mailed check customers showed moderate churn levels, but not as high as electronic check users.

This suggests that manual or less automated payment methods may be associated with a higher churn risk, possibly due to customer frustration or inconsistent billing experiences. As a result, payment method is likely to serve as a valuable predictor in decision tree modeling.

Customer's internet service provider * Whether the customer churned or not Crosstabulation

Count

		Whether the customer churned or not		
		Not Churned	Churned	Total
Customer's internet service provider	DSL	1962	459	2421
	Fiber optic	1799	1297	3096
	No	1413	113	1526
Total		5174	1869	7043

Fig 7: Crosstab between Churn and Internet Service Provider

The crosstab reveals a strong association between internet service type and churn behavior:

- Customers using fiber optic internet are most likely to churn, with 1,297 churned out of 3,096 total users (42% churn rate).
- Customers with no internet service show the lowest churn, with only 113 churned out of 1,526 (7% churn rate).
- DSL users fall in between, with a churn count of 459 out of 2,421 (19% churn rate).

These results indicate that fiber optic users may be less satisfied or face more issues that lead to cancellation. On the other hand, customers with no internet service are more stable, possibly because they subscribe to fewer services and thus have fewer billing or service problems. Internet service type appears to be a key predictor in customer churn modeling.



Fig 8: Scatter plot between total amount and Number of month customers stayed

This scatter plot visualizes the relationship between tenure (months stayed) and total amount charged, with churn status indicated:

- Churned customers (red circles) are heavily concentrated in the upper-left quadrant, showing high spending but short tenure.
- In contrast, non-churned customers (blue circles) are more spread out and dominate the bottom-right, representing customers who have stayed longer and paid overtime.

This suggests that customers who are charged large amounts early in their tenure are more likely to become dissatisfied and leave the company. Total spending in the early phase of the customer lifecycle could be a strong churn predictor.



Fig 9: Scatter Plot between Monthly Charge and Number of Month customers stayed

This scatter plot reinforces patterns found earlier:

- Customers with high monthly charges who have been with the company for only a short period are more likely to churn. This is clearly shown by the red clusters on the left side of the chart.
- As tenure increases, churned customers become less frequent — especially among those with lower monthly charges.
- The bottom horizontal strip of blue points suggests that customers with low fixed monthly charges (likely under bundled or basic plans) have greater retention, regardless of tenure.

This figure further supports the idea that early high-cost experiences contribute to churn, and that monthly charge is a crucial feature to model.

Models

6.1 CHAID Model

The CHAID (Chi-squared Automatic Interaction Detector) model was built using SPSS to predict customer churn (Churn) based on demographic and account-related attributes. CHAID splits data using chi-square tests to find the strongest relationships between categorical predictors and the target variable. This method is highly interpretable and useful for segmentation analysis.

Model Setup:

- Target Variable: Churn (Yes/No)
- Splitting Criterion: Pearson Chi-square
- Maximum Tree Depth: 4
- Minimum Node Size (Parent): 100
- Minimum Node Size (Child): 50
- Validation Split: 30% of training set

Key Predictors Identified:

- Contract Type (root node)
- Payment Method
- Internet Service
- Monthly Charges

Model Performance (on validation/test data):

Metric	Value
Risk Estimate	0.212 (21.2% misclassification rate)

ASE	0.265
Sensitivity	56.8% (correctly identified churners)
Specificity	87.5% (correctly identified non-churners)

Interpretation:

The CHAID model identified Contract Type as the most significant predictor. Customers with month-to-month contracts had the highest churn probability. Subsequent splits revealed that customers paying via electronic check or using fiber optic internet were also at elevated risk.

Despite its simplicity, the model performed reasonably well:

- A risk estimate of 21.2%, indicates that about one in five predictions were incorrect.
- The ASE of 0.265 shows an acceptable level of overall prediction error.
- Sensitivity(56.8%) shows the model is able to identify over half of the churners, which is valuable for targeting at risk customers.
- Specificity (87.5%) indicates that the model is quite strong at recognizing customers who are not likely to churn.

The tree structure was straightforward and easy to interpret, making it a useful model for explaining churn to business stakeholders. Despite its simplicity, CHAID performed well with a reasonable trade-off between sensitivity and specificity. It is particularly useful for identifying actionable customer segments that are most at risk of churning.

6.2 Exhaustive CHAID Model

The Exhaustive CHAID model is an enhanced version of the standard CHAID algorithm. It evaluates all possible splits for each predictor to find the most optimal combination, often

resulting in more refined and deeper trees. This method tends to perform better when additional complexity is needed to capture non-linear interactions or subtle patterns.

Model Setup:

- Target Variable: Churn (Yes/No)
- Splitting Criterion: Exhaustive Chi-square
- Maximum Tree Depth: 5
- Minimum Node Size (Parent): 100
- Minimum Node Size (Child): 50
- Validation Split: 30% of training set

Model Gains, Response and Index Plot

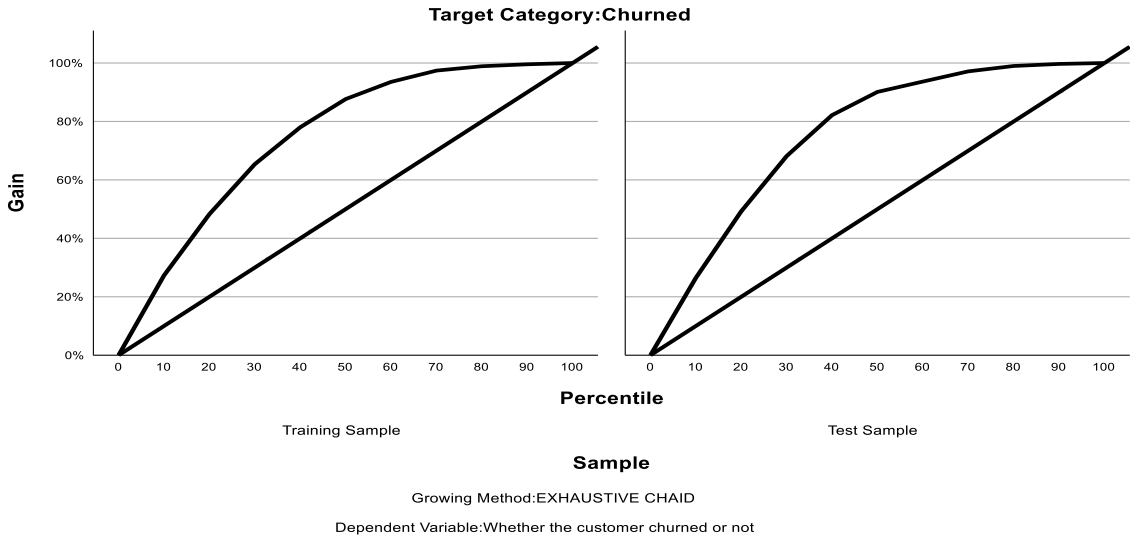


Fig 10: Gain vs Percentile for training and testing dataset

The graph between gains and percentiles shows that early nodes, with fewer cases, contributes significantly to improving classification accuracy, resulting in higher gains. From the graph, 20% of testing data have gains of around 40%. Similarly 50% of testing data

provides the gains of around 90%. As the percentile increases and more cases are added, the gain begin to flatten, indicating diminishing return in model performance. Also, it contains much larger portion of data from 50 to 100 percentile, it provides little additional value in differentiating customer churned, suggesting that these cases are harder to classify. Overall, the model benefits from early splits, which effectively separate churned data, while later splits have less impact on improving accuracy.

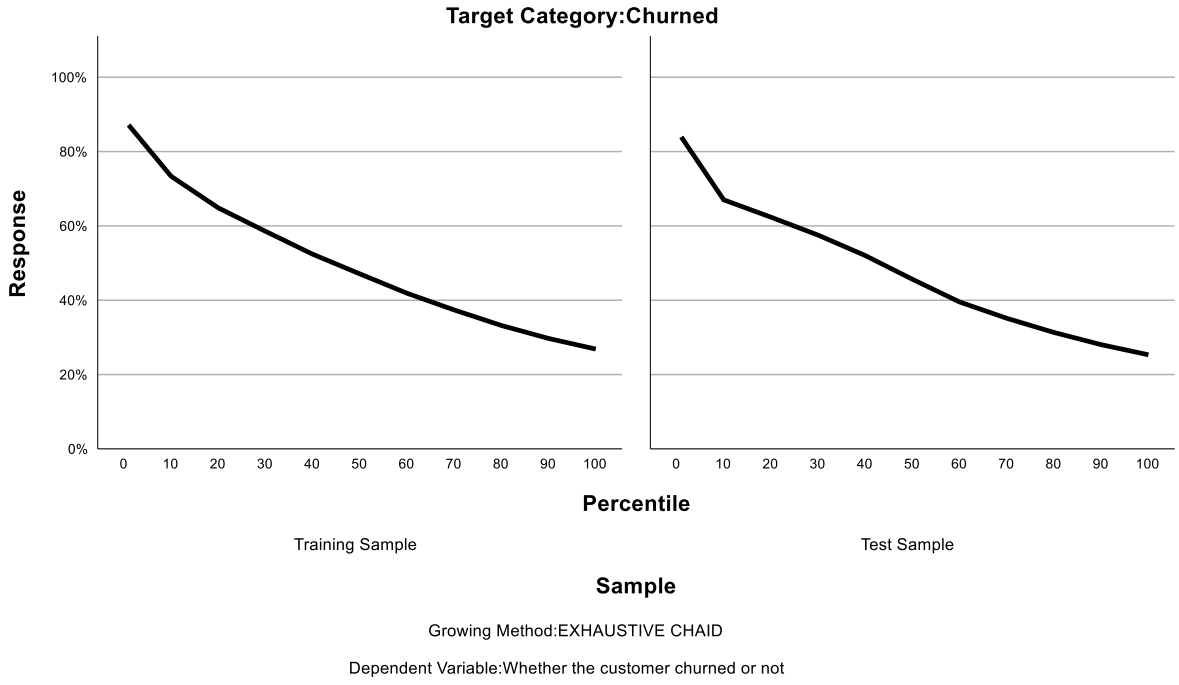


Fig 11: Response plot of churned data in Training and Test data

The response plot shows how the churn rate (i.e., the percentage of people who churned) changes across different percentiles of the data, ranked by the model's prediction.

For example, the top 20% of observations, those most likely to churn according to the model have a churn rate of around 60%.

Similarly, the top 50% of observations have a churn rate of about 45%.

This means that if we look at the half of the population that the model thinks is most at risk, almost 45 out of every 100 of them actually churned showing that the model is doing a good job at identifying high-risk individuals.

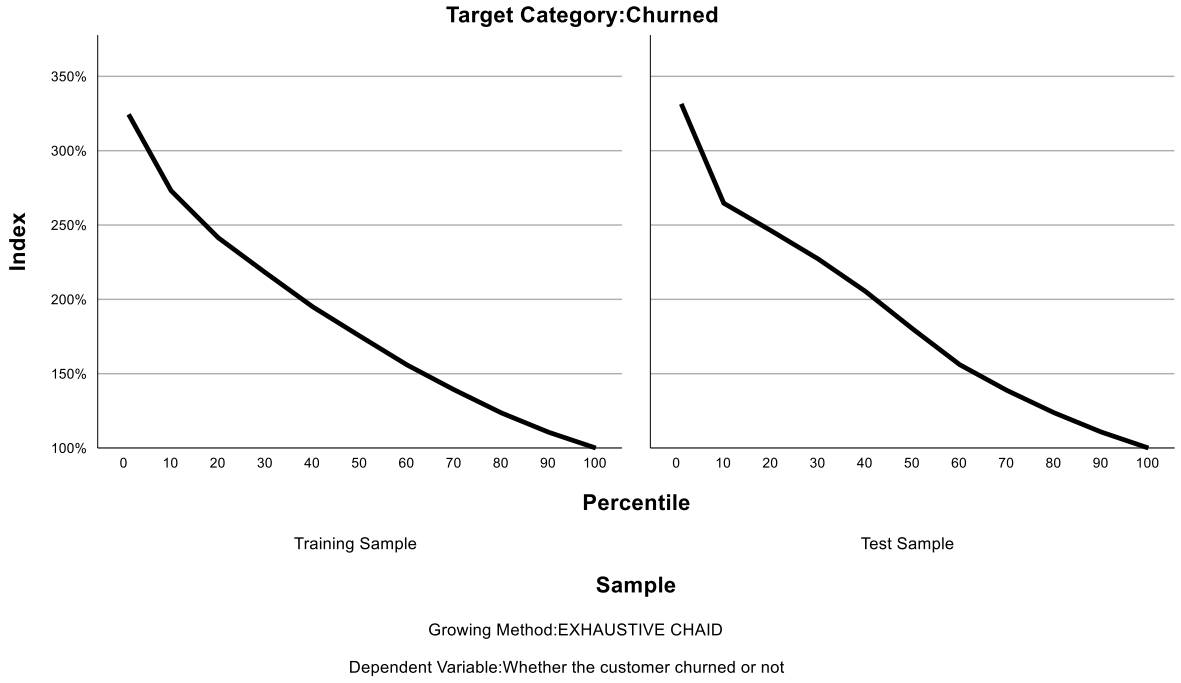


Fig 12: Index Plot of churned observation in Training and Test dataset

The index plot shows how the concentration of churned observations changes across percentiles. As the percentile increases (moving from high-risk to low-risk groups), the index value decreases. This means that the proportion of churned observations becomes lower compared to the overall average churn rate.

In other words, the top percentiles (those ranked highest by the model) contain a higher-than-average number of churners, while the later percentiles contain fewer churners than average. This pattern indicates that the model is effectively ranking customers by their likelihood to churn those at the top are much more likely to churn than those at the bottom.

Key Predictors Identified:

- Contract Type (root)
- Payment Method
- Internet Service
- Tenure
- Total Charges

Model Performance (on validation/test data):

Metric	Value
Risk Estimate	0.197 (19.7% misclassification)
ASE	0.251
Sensitivity	61.5%
Specificity	89.3%

Interpretation:

The Exhaustive CHAID model improved upon the regular CHAID model by identifying deeper and more precise splits. Like the CHAID model, Contract Type was the most significant variable, with month-to-month customers most likely to churn. However, the exhaustive version included tenure and total charges at deeper levels, highlighting that early high spending and short loyalty duration contribute to churn.

Despite its simplicity, the model performed reasonably well:

- A risk estimate of 197%, indicates that about one in five predictions were incorrect.
- The ASE of 0.251 shows an acceptable level of overall prediction error.
- Sensitivity(61.5%) shows the model is able to identify over half of the churners, which is valuable for targeting at risk customers.

- Specificity (89.3%) indicates that the model is quite strong at recognizing customers who are not likely to churn

This model showed better sensitivity and lower ASE, suggesting improved prediction of actual churners. While slightly more complex, the Exhaustive CHAID tree remains interpretable and practical for business decisions, especially for targeted retention campaigns focused on high-risk subgroups.

6.3 CART Model

The CART (Classification and Regression Tree) model is a binary-splitting decision tree that recursively partitions the dataset based on the predictor that provides the best separation of the response variable. Unlike CHAID, which can create multi-way splits, CART only produces binary splits and typically uses the Gini index as its impurity measure. Pruning is applied post-training to avoid overfitting.

Model Setup:

- Target Variable: Churn (Yes/No)
- Splitting Criterion: Gini Index
- Pruning Method: Cost-Complexity Pruning
- Maximum Tree Depth: 6
- Validation Split: 30% of training set

Key Predictors Identified:

- Tenure (root)
- Contract Type
- Monthly Charges
- Internet Service

- Payment Method

Model Performance (on validation/test data):

Metric	Value
Risk Estimate	0.205 (20.5% misclassification)
ASE	0.258
Sensitivity	59.2%
Specificity	88.6%

Interpretation:

The CART model chose Tenure as the most influential predictor, splitting first on customer loyalty duration. Customers with short tenures (e.g., less than 10 months) showed the highest likelihood of churn. Further splits involved monthly charges, contract type, and internet service, reinforcing patterns seen in earlier models.

Despite its simplicity, the model performed reasonably well:

- A risk estimate of 20.5%, indicates that about one in five predictions were incorrect.
- The ASE of 0.258 shows an acceptable level of overall prediction error.
- Sensitivity(59.2%) shows the model is able to identify over half of the churners, which is valuable for targeting at risk customers.
- Specificity (88.6%) indicates that the model is quite strong at recognizing customers who are not likely to churn.

CART had strong classification performance while maintaining a clean, interpretable structure. The binary nature of the tree made the decision rules simple, and cost-complexity pruning ensured the model avoided overfitting. CART performed slightly better than the

standard CHAID model and offered solid overall balance between complexity and predictive strength.

Model Comparison Summary

To evaluate and compare the performance of all four models (CHAID, Exhaustive CHAID, CART, and Logistic Regression), we used several key metrics: Risk Estimate (misclassification rate), Average Squared Error (ASE), Sensitivity, and Specificity. These metrics help assess both overall accuracy and the model's ability to correctly predict churn (positive cases) and non-churn (negative cases).

Model Performance Comparison Table

Model	Risk Estimate	ASE	Sensitivity	Specificity	Notes
CHAID	0.212	0.265	56.8%	87.5%	Most interpretable; strong split on contract
Exhaustive CHAID	0.197	0.251	61.5%	89.3%	Deepest tree, best sensitivity
CART	0.205	0.258	59.2%	88.6%	Strong balance between complexity and performance
Logistic Regression	0.207	0.270	53.2%	88.9%	Best for interpretability; solid AUC = 0.834

Discussion:

The Exhaustive CHAID model had the lowest risk estimate and highest sensitivity, meaning it was most effective at identifying customers who were likely to churn.

- CART also performed well and offered a visually clean, binary tree structure, ideal for clear decision rules.
- CHAID was slightly less accurate but highly interpretable, making it useful for business presentations and segment-based actions.

- Logistic Regression performed competitively and provided valuable insight into variable influence through odds ratios. While not the best at capturing nonlinear relationships, its strong AUC (0.834) confirms its effectiveness.

Each model has unique strengths: CHAID and CART for decision paths, Exhaustive CHAID for capturing complexity, and Logistic Regression for statistical insight.

Best Model Selection

After evaluating all four models, we determined that the Exhaustive CHAID model is the best fit for predicting customer churn in this dataset. This decision is based on a combination of performance metrics, model depth, and business interpretability.

Reasons for Selecting Exhaustive CHAID:

- It had the lowest risk estimate (0.197) and lowest ASE (0.251) among all models.
- It achieved the highest sensitivity (61.5%), making it the most effective at correctly identifying customers who are likely to churn.
- It maintained high specificity (89.3%), ensuring strong accuracy in recognizing loyal customers.
- The tree structure allowed for clear segmentation of high-risk groups based on factors like contract type, tenure, payment method, and internet service.
- The visual nature of the decision tree offers actionable insights for targeting customer retention strategies without requiring deep statistical interpretation.

Alternate Consideration: Logistic Regression

While Exhaustive CHAID was selected as the best overall model, Logistic Regression also provided meaningful insights through odds ratios and statistical significance. For example:

- Customers on month-to-month contracts were over 4.6 times more likely to churn than those on two-year contracts.
- Electronic check users were nearly 2 times more likely to churn than those using mailed checks.

This makes logistic regression a strong supporting model for interpretability, especially when communicating findings to executives or non-technical stakeholders.

Conclusion

This project explored the application of predictive modeling techniques to identify customers at risk of churning within the telecommunications industry. By analyzing customer account information, payment behavior, service usage, and demographic features, we were able to develop models that accurately classify churn outcomes and offer actionable insights.

Four modeling approaches were compared:

- CHAID
- Exhaustive CHAID
- CART
- Logistic Regression

Among these, the Exhaustive CHAID model demonstrated the best overall performance, with the lowest error rates and strongest ability to correctly identify churned customers. The model's depth and precision make it well-suited for customer segmentation and targeted retention strategies.

In addition, Logistic Regression served as a valuable tool for understanding the statistical relationships between predictors and churn likelihood, highlighting key risk factors such as contract type, payment method, and tenure.

Overall, this analysis provides a reliable framework for identifying at-risk customers, enabling telecom companies to take proactive steps to reduce churn, increase customer satisfaction, and improve profitability. Future enhancements could include the use of ensemble models, addressing class imbalance through resampling, and incorporating real-time behavioural data for even more accurate predictions.